

MICAH KEEGAN

669-225-4783 | micahkeegan@berkeley.edu | [linkedin.com/in/micahkeegan](https://www.linkedin.com/in/micahkeegan) | github.com/fairfax33

EDUCATION

University of California, Berkeley – College of Engineering & College of Letters and Sciences May 2028
B.S. Electrical and Computer Engineering · B.A. Physics GPA 3.809
Relevant Coursework: Physics E&M + Mechanics/Special Relativity, Computational Techniques in Physics, Program Structures, Data Structures, Machine Structures, Calc I/II/III, Lin Alg/Diff Eq, Signals & Linear Systems

PROFESSIONAL EXPERIENCE

PEAR Volunteering Santa Clara, CA
Technical Lead, Head of Outreach July 2024 – Present

- Architected & shipped full-stack platform on Next.js/React + PostgreSQL (Prisma) w/ Google OAuth for tokenized third-party verification; built RESTful API routes: opportunity geolocation, hour tracking & verification, email alerts
- Conducted stakeholder interviews w/ administrators & students to scope MVP; led 2025-26 pilot w/ 22 seniors each meeting graduation requirements; 2026-27 launch to serve 1,700+ students at net-zero cost, with goals to scale further

49ers STEM Leadership Institute Santa Clara, CA
Student Leadership Committee President Dec 2024 – May 2025

- Nominated & elected president of internal student-directed team; directly mentored in leadership by Chevron executives
- Organized 12 leaders to develop & teach curriculum to 240 academically inclined students across 6 grades; science fair & FLL mentorship increased cross-cohort engagement by 300%, led fundraisers resulting in \$760,000+ raised for program

STEM Service Learning and Mentorship San Jose, CA
Founder August 2023 – April 2025

- Volunteered 4+ hours weekly bringing STEM exposure to underserved students in-person at a Title-1 elementary school
- Grew team to 20+ volunteers who developed & taught STEM curricula to 70+ 4th & 5th graders in after-school program

Department of Psychiatry and Biobehavioral Sciences, UCLA Los Angeles, CA
Clinical Research Assistant (ClinicalTrials.gov: NCT05532683) July 2023 – May 2024

- Conducted literature review via Web of Science, PubMed, & PsycINFO to identify evidence-based behavioral strategies; managed 8-module bibliographic library in Zotero; iterated on slide content & facilitator materials with Dr. Ruiz-Yu
- Developed secondary social validity analysis on 9-person cohort, directly informing next-phase prevention programming

PROJECTS & RESEARCH

FPGA Probabilistic-Bit Ising Machine for Hardware-Accelerated Optimization February 2026 – Present

- 32-spin probabilistic-bit Ising machine in SystemVerilog on Digilent Arty A7-100T (Xilinx Artix-7) FPGA: fixed-point stochastic p-bit units with piecewise-linear tanh approximation, BRAM-backed coupling matrix, per-spin LFSR pseudorandom generators, & chromatic Gibbs sampling schedule enabling parallel spin updates without race conditions

2D Molecular Dynamics Simulation with Lennard-Jones Potential August 2025 – January 2026

- Numba-JIT-compiled physics kernels (~500x speedup), cell-list neighbor search w/ linked-list traversal ($O(N^2)$ to $O(N)$ force compute), velocity-Verlet symplectic integration, & periodic boundary conditions via minimum-image convention
- Demonstrated convergence of particle speeds to theoretical Maxwell-Boltzmann distribution from non-equilibrium initial conditions; quantified accuracy via L1, chi-squared, & max-bin error metrics (<0.05 for 500,000 particles), benchmarked CPU vs projected GPU scaling, & built ImageIO-streamed GIF pipeline for memory-efficient trajectory visualization

Nuclear Spin Initial-State Optimization for NMR Power Absorption September 2024 – May 2025

- Experimental pipeline built in Python/Anaconda using PULSEE (Brown University quantum simulation library built using QuTiP), ran 108 trials across 27 initial-state configurations (3^3 coupled-spin combinations) $\times$ 4 pulse durations: constructed Hamiltonians (Zeeman, quadrupolar, EM coupling), applied resonant RF pulses via Liouville-von Neumann evolution, generated FID signals & extracted NMR spectra via Fourier transform for quantitative density matrix analysis

ADDITIONAL INFORMATION

Primary Languages: Python, Java, C, Rust, SQL (PostgreSQL), MATLAB, SystemVerilog, LaTeX

Hardware & Fab: FPGA (Xilinx Artix-7), RISC-V, KiCAD, Fusion 360, Solidworks, Vivado, CAM/CNC, 3D Printing

Frameworks, Libraries & Tools: Next.js, React, NumPy, Numba, CUDA QuTiP, Prisma, Git, Conda, Claude Code

Interests: Soccer (Arsenal fan!), Climbing, Cooking, Reading, Fishing, Skiing, Surfing, Gym, Rocket League